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Vellore Institute of Technology

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BCSE355L - Cloud Architecture Design

**Project Title -: Reinforcement Learning-Based
Intelligent Queue Optimization Framework for
Railway Reservation Systems**

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Dataset Details

1) Dataset Name

Indian Railway Ticket Confirmation Dataset

2) Source

Kaggle

3) URL

[Indian Railway Ticket Confirmation Dataset \(Kaggle\)](#)

4) Dataset Size

Approximately 2–5 MB (CSV format).

5) Number of Records

Approximately 10,000 reservation records. *(The exact count will depend on the version downloaded, so verify it after downloading and update this value if needed.)*

6) Number of Features

Approximately 15–20 attributes, including reservation status, waiting list information, travel class, quota, train type, journey details, and booking characteristics.

7) Data Type

Structured CSV dataset containing:

- Numerical attributes
- Categorical attributes
- Reservation status labels

- Waiting-list information

8) License

MIT License, allowing educational and research use.

9) Purpose of using the dataset

The dataset will be used to simulate railway reservation scenarios and train the Reinforcement Learning model for intelligent queue optimization. Historical booking and waiting-list information will help the RL agent learn better seat allocation strategies after ticket cancellations. The dataset will also be used to evaluate system performance in terms of waiting time, seat utilization, and reservation efficiency.

10) Preprocessing Required

Before training the Reinforcement Learning model, the following preprocessing steps will be performed:

- Remove duplicate and incomplete records.
- Handle missing values in reservation attributes.
- Encode categorical features such as travel class, quota, and train type into numerical values.
- Normalize numerical attributes where required.
- Select relevant features for queue optimization.
- Split the processed dataset into training and testing data for performance evaluation.